### 5.6.7 Logarithmic Series

The series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{x^n}{n}$ is called a ***logarithmic series***. This series converges for all values of $x$ satisfying $|x|<1$. This series converges when $x=1$ also.

For all values of $x$ satisfying $|x|<1$, the sum of the series is $\log(1+x)$. Thus

$$\log(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots \text{ for all values of } x \text{ satisfying } |x|<1.$$

By taking $-x$ in place of $x$ we get

$$\log(1-x)=-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}-\cdots \text{ for all values of } x \text{ satisfying } |x|<1.$$

Now $\log\left(\frac{1+x}{1-x}\right)=\log(1+x)-\log(1-x)$. Using this we get

$$\log\left(\frac{1+x}{1-x}\right)=2\left[x+\frac{x^3}{3}+\frac{x^5}{5}+\cdots\right].$$

Suppose we want to write $\log(1+2x)$ in a series, then we can replace $2x$ by $y$ and use the expansion

$$\log(1+y)=y-\frac{y^2}{2}+\frac{y^3}{3}-\frac{y^4}{4}+\cdots$$

for all values of $y$ satisfying $|y|<1$. But if $|y|<1$, then $|2x|<1$ and hence $|x|<\frac{1}{2}$. So if $|x|<\frac{1}{2}$, then

$$\log(1+2x)=2x-\frac{(2x)^2}{2}+\frac{(2x)^3}{3}-\frac{(2x)^4}{4}+\cdots.$$

Thus $\log(1+2x)=2x-\frac{4x^2}{2}+\frac{8x^3}{3}-\frac{16x^4}{4}+\cdots$ for all values of $x$ satisfying $|x|<\frac{1}{2}$.

## Exercise - 5.4

1. Expand the following in ascending powers of $x$ and find the condition on $x$ for which the binomial expansion is valid.
   (i) $\frac{1}{5+x}$ (ii) $\frac{2}{(3+4x)^2}$ (iii) $(5+x^2)^{\frac{2}{3}}$ (iv) $(x+2)^{-\frac{2}{3}}$
2. Find $\sqrt[3]{1001}$ approximately (two decimal places).
3. Prove that $\sqrt[3]{x^3+6}-\sqrt[3]{x^3+3}$ is approximately equal to $\frac{1}{x^2}$ when $x$ is sufficiently large.
4. Prove that $\sqrt{\frac{1-x}{1+x}}$ is approximately equal to $1-x+\frac{x^2}{2}$ when $x$ is very small.
5. Write the first 6 terms of the exponential series (i) $e^{5x}$ (ii) $e^{-2x}$ (iii) $e^{\frac{1}{2}x}$.
6. Write the first 4 terms of the logarithmic series (i) $\log(1+4x)$ (ii) $\log(1-2x)$ (iii) $\log\left(\frac{1+3x}{1-3x}\right)$ (iv) $\log\left(\frac{1-2x}{1+2x}\right)$. Find the intervals on which the expansions are valid.
7. If $y=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\cdots$, then show that $x=y-\frac{y^2}{2!}+\frac{y^3}{3!}-\frac{y^4}{4!}+\cdots$.
8. If $p-q$ is small compared to either $p$ or $q$, then show that $\sqrt[n]{\frac{p}{q}}\simeq\frac{(n+1)p+(n-1)q}{(n-1)p+(n+1)q}$. Hence find $\sqrt[8]{\frac{15}{16}}$.
9. Find the coefficient of $x^4$ in the expansion of $\frac{3-4x+x^2}{e^{2x}}$.
10. Find the value of $\sum_{n=1}^{\infty}\frac{1}{2n-1}\left(\frac{1}{9^{n-1}}+\frac{1}{9^{2n-1}}\right)$.